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DATA DESCRIPTOR

Comprehensive spatiotemporal dataset of shared bicycle operations in Xiamen China

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We present a comprehensive spatiotemporal dataset of shared bicycle operations in Xiamen, China, collected over five consecutive weekdays (December 21–25, 2020) during morning peak hours (6:00–10:00 AM). The dataset comprises three components: (1) 220,675 origin-destination trip records from 53,630 unique bicycles, including precise GPS coordinates, timestamps, and lock status; (2) 2,849,243 GPS trajectory tracking points captured at 30-second to 2-minute intervals during complete trips; (3) 14,071 electronic fence records with unique identifiers and four-corner coordinates specifying parking zone boundaries. The dataset captures environmental variations including weather impacts and maintains strict privacy protection through cryptographic anonymization. This open dataset enables analysis of urban mobility patterns, cycling behavior, and demand forecasting. We demonstrate the dataset's utility through temporal distribution analysis, spatial pattern visualization, regional clustering, and demand prediction using neural networks. The dataset supports research in urban transportation planning, shared mobility optimization, and smart city applications.

Background & Summary

Shared bicycle systems have emerged as a critical component of sustainable urban transportation, providing environmentally friendly solutions to first-and-last-mile connectivity challenges in densely populated cities¹. The rapid proliferation of dockless shared bicycle systems, enabled by advances in IoT technologies and GPS tracking, has generated unprecedented volumes of spatiotemporal mobility data that provide valuable insights into urban transportation patterns and user behavior^{2,3}. China's shared bicycle market has experienced remarkable growth, with millions of shared bicycles deployed across major cities and unique operational characteristics such as dockless systems and electronic fence technologies^{4,5}.

Recent studies have demonstrated the significant potential of shared bicycle data for urban mobility analysis, with researchers leveraging large-scale datasets to investigate usage patterns^{6,7}, predict demand using machine learning approaches^{8,9} and optimize redistribution strategies¹⁰. The COVID-19 pandemic has further highlighted the importance of understanding mobility patterns in shared bicycle systems, as these systems experienced significant changes in usage due to lockdown measures and changed commuting patterns^{11,12}. Advanced analytical methods including graph neural networks and deep learning have shown particular promise in capturing spatiotemporal dependencies in mobility data¹³.

Despite the growing research interest, comprehensive datasets capturing the full operational complexity of shared bicycle systems remain limited, particularly for Asian urban contexts^{14,15}. Most existing datasets suffer from limitations including short observation periods, incomplete temporal coverage, or lack of comprehensive infrastructure information¹⁶. Widely-used open-access datasets from Western cities (e.g., Citi Bike, Santander Cycles, Capital Bikeshare) typically provide aggregated trip records from docked systems but lack continuous trajectory data and parking infrastructure details. Chinese shared bicycle data are rarely publicly available due to commercial confidentiality and privacy concerns^{17,18}. While the Shenzhen Open Data platform (<https://open-data.sz.gov.cn/data/search/toSearchPost>) provides basic trip records (start/end times, GPS coordinates, user/operator IDs¹⁹), it does not include electronic fence information, trajectory data, or validated preprocessing methodologies, limiting its analytical utility for fine-grained spatiotemporal and operational studies.

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Here, the present study is set against the backdrop of the Digital China Innovation Competition. We present a comprehensive dataset collected from Xiamen's shared bicycle system during five consecutive working days (December 21–25, 2020) from 6:00 to 10:00 AM, covering the peak morning commuting period. Xiamen is a coastal city in Fujian Province, China, with a land area of 1,699 km² and a permanent population of 5.28 million in 2020. During the study period, the shared bicycle fleet comprised around 100,000 bicycles. The dataset used in this study covers 53,630 bicycles and 220,675 valid trip records. Shared bicycles account for approximately 6–8% of weekday trips²⁰ and play a vital role in short-distance and first/last-mile travel across Xiamen's transport network. Although the dataset was collected during the COVID-19 pandemic, Xiamen had entered a low-risk stage with no active local transmission and normalized epidemic control measures at that time. Public transportation and shared bicycle services had fully resumed, and travel demand had largely returned to pre-pandemic levels.

This dataset offers significant value for multiple research domains including urban mobility analysis, transportation planning, and smart city applications. The comprehensive temporal coverage enables investigation of fine-grained usage fluctuations during critical morning rush hours, while the spatial precision supports detailed analysis of origin-destination patterns and the relationship between infrastructure distribution and usage intensity^{21,22}. The methodological framework accompanying the dataset includes validated approaches for temporal analysis, spatial distribution mapping, regional clustering, and demand prediction, demonstrating high accuracy in pattern validation and confirming the dataset's reliability for advanced analytical applications^{23,24}. Compared to existing open-access datasets, this dataset offers unique advantages through its multi-layered architecture integrating trip records (220,675 OD pairs), continuous trajectories (2,849,243 GPS points), and infrastructure data (14,071 electronic fences), it addresses critical gaps in Asian urban mobility research.

The dataset addresses critical gaps in publicly available shared bicycle data by providing comprehensive operational and infrastructure information from an Asian urban context, high temporal resolution during peak usage periods, and validated analytical methodologies. This resource supports advancing understanding of urban shared bicycle systems and developing data-driven solutions for sustainable urban transportation in the era of smart cities.

Methods

Data sources. We utilized real-time monitoring systems from major shared bicycle operators in Xiamen to gather operational data. The participating shared bicycle systems are equipped with advanced IoT infrastructure featuring dual-mode BeiDou/GPS positioning smart locks with meter-level accuracy, 4G communication modules, and multi-sensor fusion status monitoring devices. Data reporting intervals are dynamically adjusted (30 seconds to 2 minutes) based on bicycle usage status, while operational platforms aggregate real-time information including vehicle location, lock status, and fault alerts through standardized API interfaces.

Data collection period and time selection. The data were collected from December 21 to 25, 2020 (Monday to Friday), during the morning commuting hours of 6:00–10:00 a.m. each day. This period was selected to capture the typical weekday commuting behavior of shared bicycle users in Xiamen while avoiding the irregular travel patterns commonly observed on weekends and holidays.

Data structure composition. This study utilizes a multidimensional dataset that integrates three primary data sources - trip order records, trajectory data, and electronic fence information - supplemented by contextual variables such as weather conditions and street network data. The dataset covers five weekdays during morning commuting hours (6:00–10:00), providing detailed spatial and behavioral information suitable for short-term peak-hour analysis. Although the temporal scope is limited, the data's high spatial granularity and multi-source structure enable comprehensive examination of mobility dynamics, demand patterns, and operational efficiency within the study area. Specifically:

- (1) **Shared Bicycle Order Data.** The primary order dataset contains 220,675 original records, each record represents an individual bicycle status change event with the following standardized field structure:
 - BICYCLE_ID:** Unique alphanumeric identifier for each bicycle in the shared system.
 - LATITUDE:** Geographical latitude coordinate in decimal degrees (6 decimal places precision, WGS84 coordinate system)
 - LONGITUDE:** Geographical longitude coordinate in decimal degrees (6 decimal places precision, WGS84 coordinate system)
 - LOCK_STATUS:** Binary indicator of bicycle lock state (0 = unlocked/trip start, 1 = locked/trip end)
 - UPDATE_TIME:** Calendar date of the recorded event (YYYY/MM/DD format)
 - TIME:** Time of day for the recorded event (HH:MM:SS format, 24-hour clock)
 Table 1 presents a representative sample of the processed data showing sequential unlock and lock events for BICYCLE_1 on December 22, 2020, demonstrating the typical data structure and the sequence of events that constitute individual trips.
- (2) **Trajectory data.** The trajectory dataset contains 2,849,243 GPS tracking points recorded during complete bicycle trips. Each tracking point includes:
 - Bicycle identifier (anonymized)
 - Precise GPS coordinates (latitude/longitude, WGS84)
 - Timestamp (synchronized with order data)
 - Recording interval (30 seconds to 2 minutes based on usage status)

BICYCLE_ID	LATITUDE	LONGITUDE	LOCK_STATUS	UPDATE_TIME	TIME
BICYCLE_1	24.509607	118.155216	0	2020/12/22	8:00:44
	24.494801	118.147564	1	2020/12/22	8:14:57
	24.494648	118.147526	0	2020/12/22	8:19:35
	24.491365	118.149005	1	2020/12/22	8:23:24
	24.491301	118.149002	0	2020/12/22	8:50:25
	24.493236	118.141339	1	2020/12/22	8:54:24
	24.493205	118.141015	0	2020/12/22	8:55:23
	24.498513	118.14314	1	2020/12/22	8:59:03
	24.499019	118.142731	0	2020/12/22	9:07:42
	24.496973	118.142227	1	2020/12/22	9:09:56
	24.495896	118.141908	0	2020/12/22	9:12:04
	24.494854	118.130241	1	2020/12/22	9:22:09

Table 1. Order data with origin-destination pairs.



Fig. 1 Visual visualization of order data.

Figure 1 provides a comprehensive visualization of these trip data, illustrating the spatial and temporal patterns captured in the individual bicycle usage records.

- (3) **Electronic Fence Data.** The electronic fence dataset contains 10522 records, each representing a rectangular authorized parking area for shared bicycles within Xiamen’s urban area. Each record includes FENCE_ID (unique identifier for the parking area) and FENCE_LOC (coordinate sequence defining the four corner points of the rectangular parking area), with the coordinate data structure containing latitude and longitude values for all four corners of each rectangular parking zone, stored as complete polygonal boundaries that accurately reflect the actual geometric shape of designated parking areas in the urban environment.
- (4) **Road network infrastructure.** To address spatial analysis requirements, we obtained comprehensive road network data from OpenStreetMap²⁵ (OSM) for the Xiamen region (<https://download.geofabrik.de/asia/china/fujian.html>). The road network dataset was systematically filtered to include bicycle-accessible infrastructure: bicycle-specific infrastructure, mixed-traffic roads accessible to bicycles, and pedestrian infrastructure relevant to shared bicycle. The raw OSM road network underwent systematic processing, removing highway segments restricted to motor vehicles, excluding private roads, and retaining service roads and mixed-use pathways commonly utilized by shared bicycle users.
- (5) **Spatial analysis and heatmap generation.** Spatial visualization and map-based trajectory analysis were conducted using the Dithubui²⁶ mapping platform (<https://e.dithubui.com/basedata>), an online geographic information system (GIS) service that supports multi-layer data visualization, coordinate projection, and path rendering. The platform enables interactive visualization of trajectory data, electronic fence boundaries, and spatial distribution patterns of shared bicycles within Xiamen.
- (6) **Weather data.** Meteorological data²⁷ with hourly resolution for December 21–25, 2020, including: Temperature (°C), Precipitation (mm), Wind speed (km/h), Humidity (%), Visibility (km). Source: (<https://meteo-stat.net/en/station/59134?t=2020-12-20/2020-12-25>).

Order duration (s)	Data size	Data screening rate	Valid data rate
<60	6301	2.86%	—
[60, 3600]	212630	2.88%	97.12%
[120, 3600]	210007	4.87%	95.13%
[180, 3600]	196394	11.09%	88.91%
[240, 3600]	174749	20.98%	79.02%
[300, 3600]	150595	32.01%	67.99%
>3600	1744	0.79%	—

Table 2. Results of data preprocessing showing the impact of different order duration thresholds.

Data filtering statistics. Order duration filtering. Order duration filtering refers to the process of identifying and removing trip records with abnormally short durations that do not represent actual riding behavior but rather reflect system malfunctions, accidental unlock-lock events, or maintenance operations. These anomalous records were excluded from the dataset by applying a minimum trip duration threshold, ensuring that only valid usage data were retained.

Common sources of abnormal order durations include:

- (1) accidental unlock-lock sequences where users re-lock bicycles without riding;
- (2) internal system testing or maintenance operations;
- (3) GPS signal latency leading to inconsistent timestamps;
- (4) hardware malfunctions in smart lock systems.

Data preprocessing. To improve data quality and ensure the reliability of mobility pattern analysis, we applied a duration-based filtering strategy. Trip records with durations shorter than 60 seconds or longer than 3600 seconds were excluded. Extremely short trips (<60 s) are typically caused by accidental unlocks, smart lock malfunctions, or system testing operations, and do not represent actual bicycle usage. Similarly, trips exceeding one hour (>3600 s) are rare and often result from abnormal conditions such as improper locking, GPS signal loss, or manual movement without registration.

Table 2 summarizes the impact of different order duration thresholds on data volume and validity. Records within the [60, 3600] second range account for 212,630 trips, representing 97.12% of all valid data. Only 0.79% of trips exceed one hour (1,744 records), further supporting the assumption that long-duration trips are outliers.

Stricter lower-bound thresholds (e.g., [120, 3600], [180, 3600], etc.) result in a significant reduction in data volume and valid rate—from 95.13% at 120 seconds down to 67.99% at 300 seconds. This pattern demonstrates that many genuine short trips would be lost if more aggressive filtering were applied. Thus, the [60, 3600] threshold was selected as the optimal balance, preserving valid short trips while removing abnormal durations at both ends.

Privacy protection. Strict privacy protection protocols were implemented throughout the data collection and processing pipeline. All bicycle IDs underwent cryptographic hashing, and user-related personal identifiers were completely removed to ensure full anonymization. The data collection, storage, and usage procedures comply with academic research ethics requirements and relevant regulatory standards.

Temporal analysis methods. We conducted comprehensive temporal analysis by aggregating bicycle unlocking events at multiple time scales to reveal usage patterns ranging from fine-grained variations to broader temporal structures. The temporal analysis methodology employed a hierarchical approach to capture both intra-day (30-minute interval) variations within the morning commuting period and inter-day patterns across the five consecutive weekdays. This allowed us to examine fine-grained operational dynamics as well as day-to-day fluctuations in shared bicycle usage.

First, we examined overall usage patterns across the complete five-day observation period to identify day-to-day consistency and variations in usage levels. This analysis provided insights into the stability of usage patterns and external factors that might influence day-specific variations.

Subsequently, we conducted detailed temporal analysis by systematically dividing each four-hour morning observation period (6:00–10:00) into 30-minute intervals. This fine-grained temporal resolution enables observation of precise usage fluctuations and identification of specific peak periods that are critical for operational planning and resource allocation decisions.

Spatial analysis methods. For comprehensive spatial distribution analysis, we utilized the precise unlocking and locking positions recorded in the order data as origin and destination points for individual trips. This approach provides detailed insights into spatial usage patterns, popular routes, and the relationship between origins and destinations across the urban landscape.

The heatmaps of shared bicycle usage were generated through Dituhui's kernel density estimation (KDE)-based visualization tool, which applies a Gaussian kernel function to calculate the spatial density of trip origins and destinations. The resulting heatmap intensity represents the relative concentration of shared bicycle activities, where warmer colors indicate higher levels of usage.

Regional division and clustering. To facilitate operational management and optimize bicycle redistribution strategies, we developed a sophisticated regional division approach based on two fundamental factors that directly impact operational efficiency and user experience:

Distance calculation. The walking distance between electronic fence locations, calculated using the enhanced formula:

$$d_{ij} = \omega \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \quad (1)$$

where ω represents a conversion coefficient determined by the complexity of the actual road network infrastructure, and x, y denote the precise coordinates of electronic fence locations.

Spatial analysis and heatmap generation. In the context of shared bicycle systems, demand complementarity quantifies the degree to which regions exhibit mutually balancing demand patterns that facilitate natural bicycle redistribution. Within each time interval t , the net demand of region i is defined as:

$$E_i(t) = O_i(t) - D_i(t) \quad (2)$$

where O_i and D_i denote the numbers of unlock (pickup) and lock (return) events, respectively. A positive $E_i(t)$ indicates a region with high user demand or bicycle shortage, while a negative $E_i(t)$ indicates bicycle surplus.

The demand complementarity between two regions i and j is then derived from the correlation of their net demand time series, where a stronger negative correlation reflects a higher level of complementarity and better operational synergy.

we compute the Pearson correlation $\rho_{ij}(t)$ between $E_i(t)$ and $E_j(t)$, and map it to a complementarity score $S_{ij}(t) = \max(0, -\rho_{ij}(t))$. Geographical proximity is modeled by a spatial kernel $K(d_{ij}) = \exp(-d_{ij}/\lambda)$. To account for activity intensity, we introduce a magnitude factor $M_{ij}(t) = 2 \min(|\bar{D}_i|, |\bar{D}_j|)/(\bar{D}_i + \bar{D}_j)$. The final regional affinity is defined as:

$$R_{ij}(t) = [S_{ij}(t)]^\gamma \cdot K_{ij}[d_{ij}] \cdot M_{ij}(t) \in [0, 1] \quad (3)$$

where γ is a weighting exponent that controls the influence of demand complementarity in the overall affinity measure.

For the regional clustering implementation, we employed an enhanced K-Means++ algorithm with the following systematic steps:

Initialization. Place all electronic fence nodes in set M , calculate correlation degrees for all nodes according to Eq. 3, and select the node with the highest correlation degree as the initial cluster center, designated as C_1 .

Progressive Center Selection. Calculate the shortest distance between each electronic fence node and the current cluster centers, and compute its operational correlation using Eq. (3), which is derived from the region's unlock and lock event counts. Then, assign a probability to each node based on a hybrid score proportional to distance $\times$ (1 - correlation). Nodes that are both geographically distant and operationally distinct are given higher selection probabilities. The node with the highest probability is selected as the next cluster center.

Center determination. To determine the optimal number of cluster centers (k), we conducted a systematic analysis using both the Elbow Method and the Silhouette Score. The Elbow Method evaluates the total within-cluster sum of squared errors (SSE) as a function of k , identifying the point at which additional clusters provide diminishing returns. Meanwhile, the Silhouette Score measures how similar each point is to its own cluster compared to other clusters, offering a consistency-based perspective on clustering quality. By jointly analyzing both metrics across a range of candidate k values, we selected the value of k that achieves a balance between low intra-cluster variance and high inter-cluster separation, repeating until all cluster centers are established based on correlation strength and geographical distribution.

Node assignment. Assign each node in set M to its nearest cluster center C_k , ensuring optimal grouping based on both geographical proximity and operational compatibility.

Convergence verification. Evaluate convergence by checking whether the sum of squared distances D between each electronic fence node and its nearest cluster center C_k stabilizes. If convergence is achieved, return final cluster centers $C_1, C_2, \dots, C_k$ and terminate the algorithm; otherwise, return to Step 2 for further iteration.

Demand prediction. We developed a Back Propagation Neural Network (BPNN)-based demand prediction model to forecast shared bicycle demand across different regions and time periods. The overall workflow of the

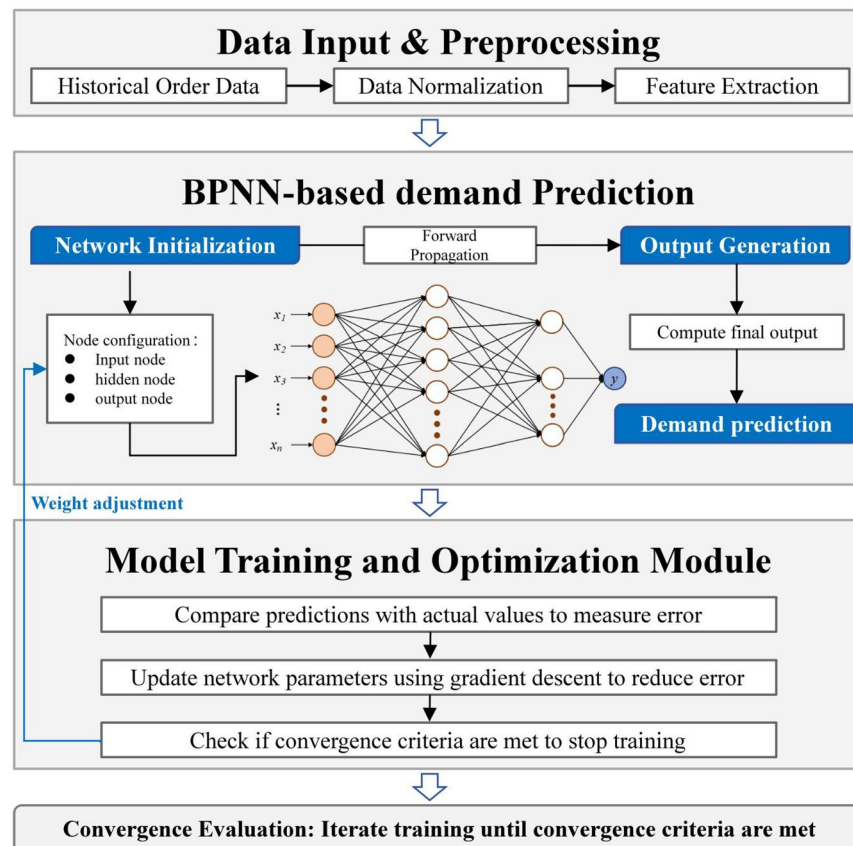


Fig. 2 Flow diagram of the Back Propagation Neural Network (BPNN)-based demand prediction model.

model is illustrated in Fig. 2, which summarizes the sequential process including data preprocessing and feature extraction, network initialization, forward and backward propagation, and model convergence evaluation. The BPNN uses multiple layers optimized for spatiotemporal prediction tasks, where forward propagation generates demand forecasts and backward propagation adjusts model weights based on prediction errors until convergence.

Data Records

The dataset is permanently archived in the Zenodo open-access repository (<https://doi.org/10.5281/zenodo.17415891>²⁸) and provided in standardized CSV (comma-separated values) format with UTF-8 encoding to ensure broad compatibility across analytical platforms. All timestamps use UTC + 8 time zone, and coordinates follow the WGS84 coordinate system.

Repository organization. The repository contains six top-level folders with comprehensive documentation:

- Order data: Trip records and processed origin-destination pairs
- Trajectory data: Continuous GPS tracking points
- Electronic fence data: Authorized parking zone boundaries
- Algorithm data: Clustering and prediction model inputs/outputs
- Data cleaning algorithm: Scripts and procedures for preprocessing
- Weather data: Meteorological conditions during study period

A README file and data dictionary document folder layout, file naming conventions, variable definitions, units, time zone specifications, and encoding details.

Technical Validation

We validated the dataset and analytical methodologies through multiple independent approaches to ensure reliability, accuracy, and meaningful interpretation of the observed patterns.

Temporal pattern validation. The usage distribution of shared bicycles on weekdays exhibits a certain degree of balance; however, demand fluctuations are significant due to environmental influences. Under unfavorable riding conditions, user demand for shared bicycles decreases compared to normal conditions. Weather data from the Xiamen Meteorological Bureau indicate that on December 23, when light rain occurred during morning peak hours, the total number of trips was approximately 60% lower than the average of other

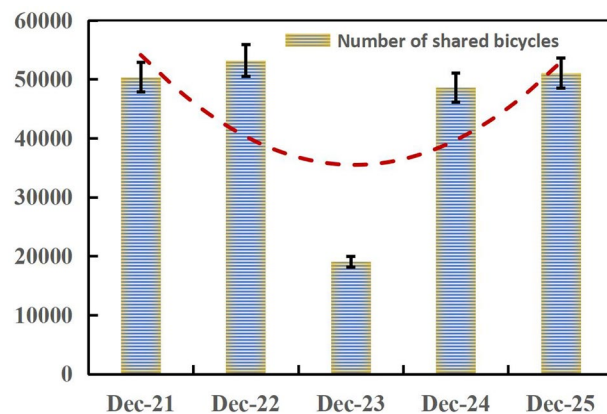


Fig. 3 Daily trip statistics of shared bicycle operations during the study period (December 21–25, 2020). The bar chart shows the total number of trips recorded each day during morning peak hours (6:00–10:00 AM). December 23 experienced light rain during data collection.

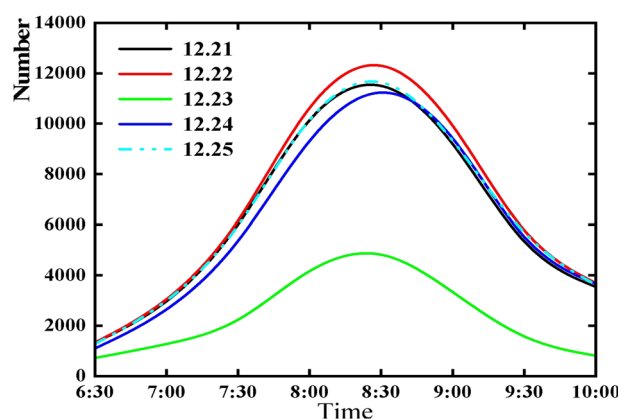


Fig. 4 Temporal distribution of shared bicycle usage across the study period. Hourly trip counts during morning peak hours (6:30–10:00 AM) for five consecutive weekdays (December 21–25, 2020). Each line represents one day, showing the characteristic peak usage pattern centered around 8:00–9:00 AM.

clear-weather weekdays. This finding confirms that precipitation significantly suppresses shared bicycle usage due to reduced riding comfort and safety. This is reflected in the daily user base, where user dependency on shared bicycles is less than 40% (see Fig. 3 for overall usage statistics).

The temporal analysis revealed highly consistent patterns across all working days in the observation period, with bicycle usage following a characteristic distribution pattern centered around 8:00–9:00 AM (as shown in Fig. 4). This temporal pattern aligns closely with expected commuting behaviors during weekday mornings and demonstrates remarkable consistency across the dataset despite variations in daily environmental conditions. The consistency of these temporal trends across multiple days confirms the reliability and representativeness of the observed usage patterns.

Cross-day correlation analysis of hourly usage patterns yields consistently high correlation coefficients (exceeding 0.90 for all day pairs), confirming the stability and predictability of temporal usage patterns. This high temporal consistency validates the dataset's suitability for developing reliable prediction models and supports its use in operational planning applications.

Spatial distribution validation. The spatial heat map clearly demonstrates logical and interpretable distribution patterns, with high usage density concentrated around key urban features such as residential areas, commercial districts, and major transportation hubs. In contrast, park areas and forested regions show relatively low usage, particularly during working day morning hours. As shown in Fig. 5, these patterns are entirely consistent with urban mobility theory and align well with Xiamen's geographical characteristics and urban development patterns. The figure also includes an overlaid administrative district boundary layer to provide clearer spatial context and improve interpretability of the clustering analysis.

To further investigate the temporal dynamics of shared bicycle usage within the selected region (as delineated by the red box in Fig. 5), a series of fine-grained spatiotemporal heat maps were generated. These heat maps illustrate the evolution of usage intensity from 6:00 AM to 10:00 AM, segmented into eight 30-minute intervals. As shown in Fig. 6, the distribution patterns clearly reveal a gradual increase in demand during early morning

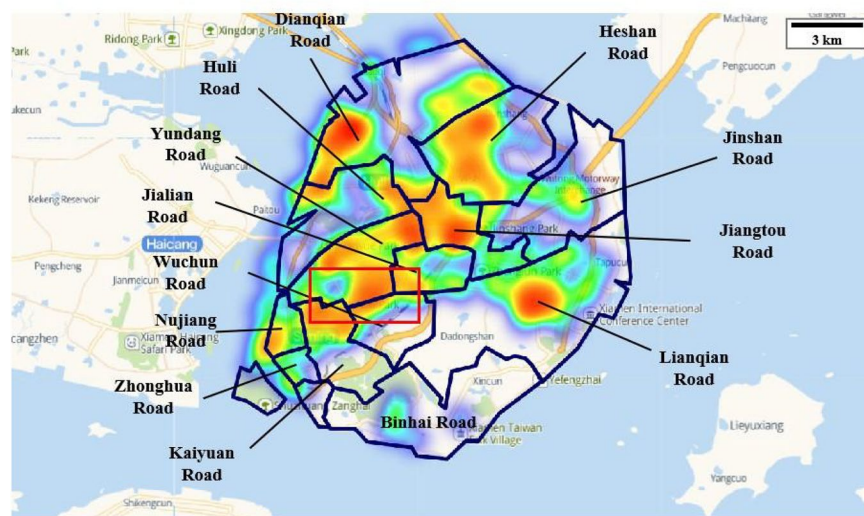


Fig. 5 Spatial heat map of shared bicycle usage with overlaid administrative district boundaries in Huli and Siming Districts, Xiamen. The red box indicates the geographical range of the analyzed area.

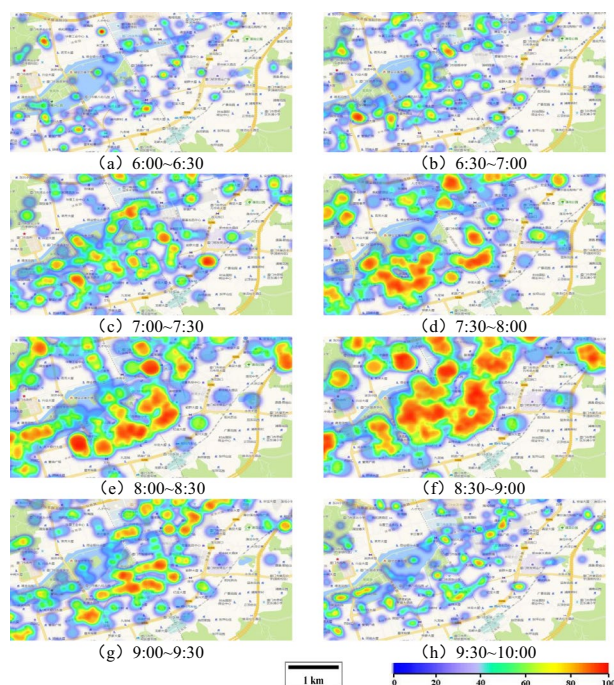


Fig. 6 Spatiotemporal heat maps of shared bicycle usage within the analyzed region (as indicated by the red box in Fig. 5), from 6:00 AM to 10:00 AM at 30-minute intervals (eight subplots in total).

hours, peaking between 8:30 AM and 9:00 AM. This temporal evolution aligns closely with typical urban commuting behaviors and provides valuable insights into the spatial diffusion of morning peak demand within the analyzed area.

Regional clustering validation. The regional clustering algorithm underwent rigorous testing using a subset of order data from December 21, 2020, during the 6:00–10:00 AM period within a specific geographical area of Xiamen (latitude range [24.461068, 24.47782], longitude range [118.086461, 118.109601]). Each coordinate point represents an individual order record (As shown in Fig. 7).

We established the conversion coefficient between actual dispatch distance and straight-line distance at 1.5²⁹, reflecting real-world road network constraints. The algorithm systematically analyzed correlation strength between nodes, arranged all node correlations from highest to lowest, and performed clustering according to correlation strength, gradually dividing regions until complete coverage was achieved.

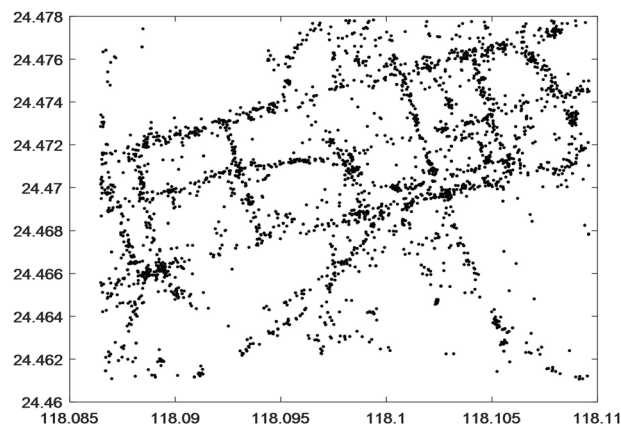


Fig. 7 Order coordinates. Geographic extent: 24.461–24.478°N, 118.086–118.110°E.

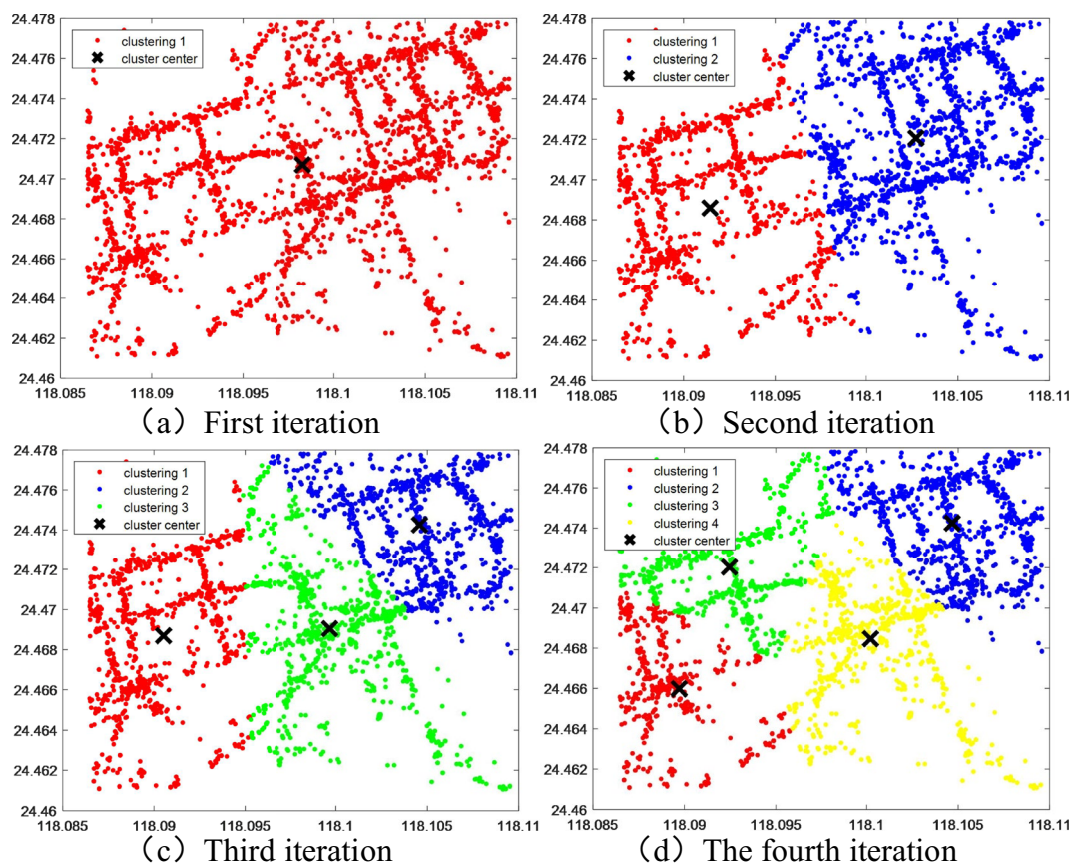


Fig. 8 Regional clustering results using the enhanced K-Means++ algorithm. The map shows the iterative progression of cluster formation based on demand complementarity and geographical proximity. Different colors represent distinct operational regions identified through the clustering process.

Figure 8 illustrates the iterative clustering process across multiple iterations, demonstrating how the algorithm successfully divided the study area into coherent operational regions based on usage patterns and geographical proximity. The clustering results show clear geographical coherence and operational logic, with each identified region exhibiting appropriate size, demand balance, and accessibility characteristics for practical bicycle management and redistribution operations.

After multiple iterations, all nodes successfully aggregated into four distinct dispatch regions. For detailed analysis, we selected the area within latitude range [24.468, 24.478] and longitude range [118.098, 118.11], corresponding to the blue region in the upper right of the fourth iteration result, as the primary research focus for model validation and algorithm testing.

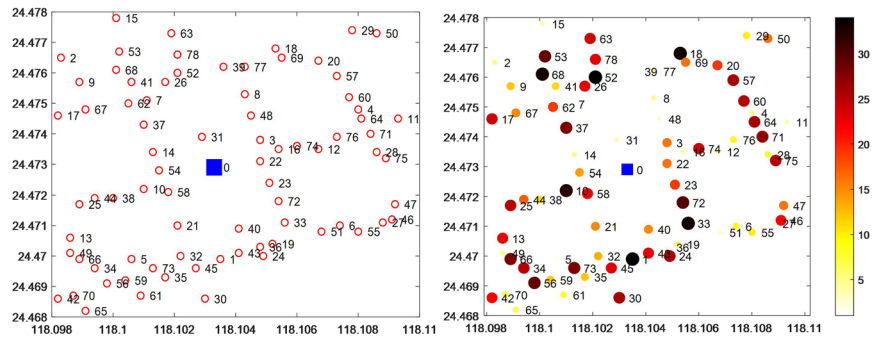


Fig. 9 Spatial distribution of electronic fence nodes based on demand characteristics. Both dot size and color depth indicate unlock frequency (demand intensity): larger and darker dots represent higher demand levels.

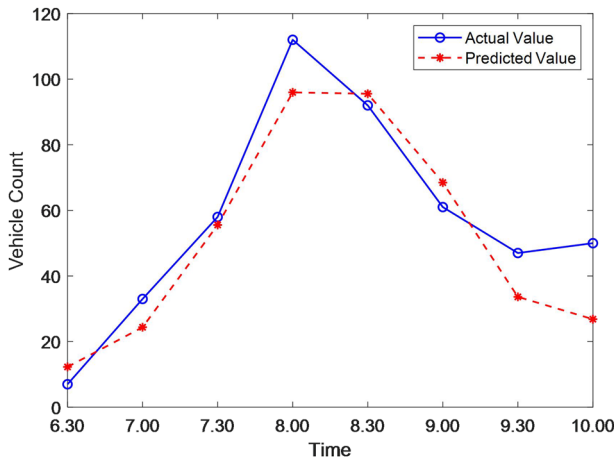


Fig. 10 BP neural network prediction results.

Date	Area A						Area D					
Time	21	22	23	24	25	...	21	22	23	24	25	
6.00~6.30	33	30	13	25	22		9	10	5	9	7	
6.30~7.00	62	61	31	56	74	·	35	21	11	28	33	
7.00~7.30	123	141	35	115	128	·	76	56	22	63	58	
7.30~8.00	301	314	137	240	297	·	101	92	45	92	112	
8.00~8.30	284	306	126	297	290	·	102	106	47	87	92	
8.30~9.00	205	205	75	185	212	·	67	76	15	76	61	
9.00~9.30	151	151	25	121	166	·	30	54	13	34	47	
9.30~10.00	161	121	43	140	136		26	23	7	31	50	

Table 3. Regional cycling data.

Given the high density of shared bicycle electronic fence nodes (418 total) within the selected data area and the relatively short distances between adjacent electronic fences, we conducted additional cluster analysis on the electronic fence nodes based on demand patterns and spatial proximity. The results were mapped to the geographic coordinate system and rendered according to site-specific demand characteristics(as show in Fig. 9), where both dot size and color intensity represent the unlock frequency at each electronic fence node, reflecting the spatial variation in bicycle demand intensity.

Demand prediction validation. We conducted comprehensive validation of the demand prediction model using a representative test case from December 25, 2020, for a selected geographical region. The model training utilized data from other regions and dates to ensure independent validation and assess generalizability of the predictive approach. The demand for bike rentals and returns can be analyzed separately. Taking the bike rental demand as an example, we aim to forecast the rental demand for December 25, 2020. Focusing primarily on the

Serial Number	Longitude	Latitude	Borrowing Demand	Returning Demand	Serial Number	Longitude	Latitude	Borrowing Demand	Returning Demand
1	118.1035	24.4699	34	9	40	118.1041	24.4709	15	5
2	118.0983	24.4765	5	9	41	118.1006	24.4757	11	5
3	118.1048	24.4738	16	2	42	118.0982	24.4686	22	5
4	118.108	24.4748	5	4	43	118.1041	24.4701	23	8
5	118.1006	24.4699	2	8	44	118.0994	24.4719	17	2
6	118.1074	24.471	9	9	45	118.1027	24.4696	23	0
7	118.1011	24.4751	2	3	46	118.1091	24.4712	22	8
8	118.1043	24.4753	5	3	47	118.1092	24.4717	16	2
9	118.0989	24.4757	12	2	48	118.1045	24.4746	3	7
10	118.101	24.4722	32	2	49	118.0986	24.4701	5	3
11	118.1093	24.4745	4	6	50	118.1086	24.4773	16	1
12	118.1067	24.4735	5	2	51	118.1068	24.4708	3	4
13	118.0986	24.4706	23	8	52	118.1021	24.476	34	10
14	118.1013	24.4734	5	2	53	118.1002	24.4767	29	5
15	118.1001	24.4778	4	0	54	118.1015	24.4728	14	9
16	118.1054	24.4735	4	2	55	118.108	24.4708	10	10
17	118.0982	24.4746	24	5	56	118.0998	24.4691	29	2
18	118.1053	24.4768	33	2	57	118.1073	24.4759	26	1
19	118.1052	24.4704	7	2	58	118.1018	24.4721	23	0
20	118.1067	24.4764	19	8	59	118.1004	24.4692	12	7
21	118.1021	24.471	15	4	60	118.1077	24.4752	25	6
22	118.1048	24.4731	16	4	61	118.1009	24.4687	8	1
23	118.1051	24.4724	19	8	62	118.1005	24.475	19	5
24	118.1049	24.47	26	5	63	118.1019	24.4773	24	3
25	118.0989	24.4717	25	5	64	118.1081	24.4745	26	5
26	118.1017	24.4757	23	1	65	118.0991	24.4682	7	3
27	118.1088	24.4711	3	7	66	118.0989	24.4699	28	0
28	118.1086	24.4734	8	6	67	118.0991	24.4748	15	5
29	118.1078	24.4774	10	5	68	118.1001	24.4761	33	1
30	118.103	24.4686	26	4	69	118.1055	24.4765	16	5
31	118.1029	24.4739	4	1	70	118.0987	24.4687	8	3
32	118.1022	24.47	13	3	71	118.1084	24.474	27	0
33	118.1056	24.4711	33	3	72	118.1054	24.4718	30	4
34	118.0994	24.4696	25	17	73	118.1013	24.4696	28	0
35	118.1017	24.4693	12	5	74	118.106	24.4736	24	2
36	118.1048	24.4703	2	2	75	118.1089	24.4732	25	8
37	118.101	24.4743	28	2	76	118.1073	24.4739	9	9
38	118.1	24.4719	11	6	77	118.1043	24.4762	5	9
39	118.1036	24.4762	1	4	78	118.1021	24.4766	23	6

Table 4. Base value of demand forecast.

blue region depicted in Fig. 6(d) and utilizing data from the other three regions as training samples, we proceed with the following steps.

Initially, we organize and select data by considering the bike rental data from December 21 to December 25, 6:00-10:00 AM, from the three non-primary regions as input variables. The output variable is the number of bike rentals in the target region on December 25. Given that the time interval is divided into 30-minute units, the test sample size is set to 8, and the training sample size is 152. A portion of the input data is presented in Table 3.

We use the `trainlm` function to train the network, with the `transig` function and the `purelin` function as the transfer functions for the hidden layer and output layer, respectively. The output network learning rate is set to 0.01, with 1000 training iterations and a minimum training error target of 0.000001. All other settings are left at the default values for the BP neural network.

After several iterations of training, the training goal is achieved. The output results are shown in Fig. 10. If the training data samples are further increased, the error range can be further minimized. After several iterations, the training goal is reached, and the specific output results are presented in Table 4.

Fig. 11 demonstrates that the prediction results closely matched actual observed demand patterns. The correlation coefficients between predicted and actual values achieved high accuracy levels: 0.99574 for training data, 0.98874 for validation data, 0.97918 for independent test data, and 0.98903 overall. These exceptional correlation values demonstrate both the high accuracy of the prediction model and, by extension, the reliability and consistency of the underlying dataset.

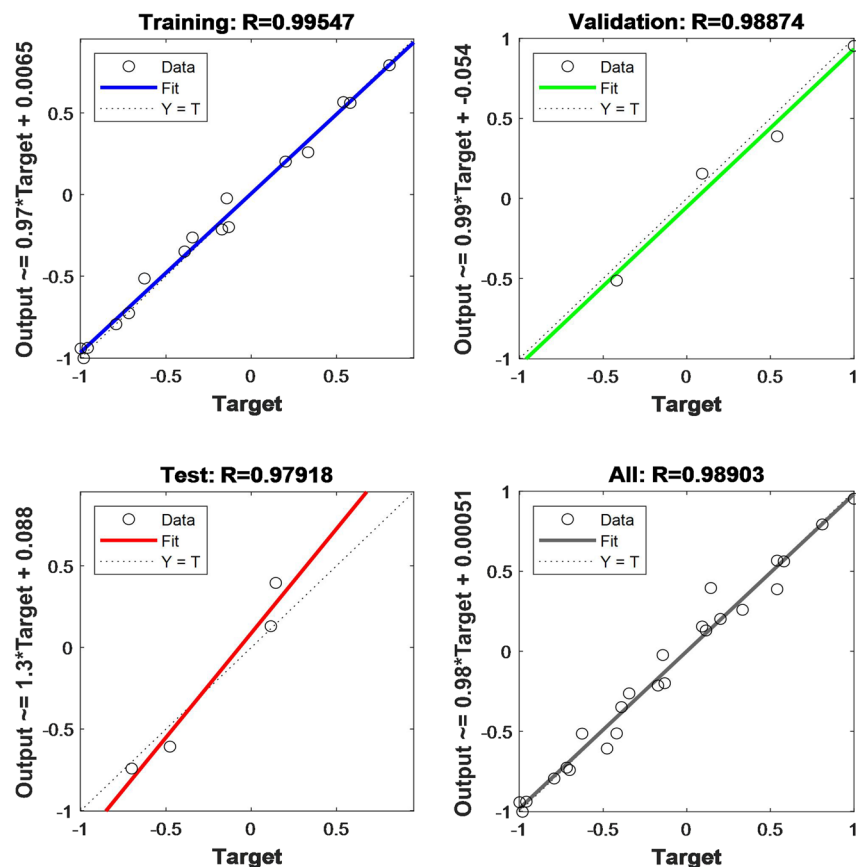


Fig. 11 Algorithm fit.

Usage Notes

This dataset provides a comprehensive view of shared bicycle operations through multi-source integration of trip records (220,675 OD pairs), continuous trajectory data (2,849,243 GPS points), and infrastructure information (14,071 electronic fences). Researchers should note several important considerations when using this dataset. First, the temporal coverage is limited to five weekday mornings (6:00-10:00 AM), providing robust insights into commuting patterns but limited information on evening peaks, weekends, or seasonal variations. Second, all coordinates use the WGS84 coordinate system (UTC + 8 time zone), and bicycle IDs are cryptographically hashed for privacy protection, preventing longitudinal tracking of individual users across days. Third, while the dataset includes comprehensive electronic fence infrastructure data, actual parking compliance rates and enforcement mechanisms are not documented. The data can be effectively integrated with complementary sources including OSM road networks (provided), public transit schedules, and census data to enable comprehensive urban mobility analysis.

Mobility pattern analysis. Although the dataset covers only five working days, it provides valuable short-term insights into weekday spatiotemporal patterns of shared bicycle usage, enabling better understanding of urban mobility behaviors, identification of popular routes and destinations, and examination of how environmental factors influence bicycle usage decisions. Our analysis demonstrates that adverse environmental conditions can reduce demand by approximately 60%, indicating that roughly 40% of users maintain consistent bicycle usage regardless of weather conditions, suggesting a core group of highly dependent users alongside more weather-sensitive occasional users.

The dataset supports investigation of daily and hourly variation patterns, peak period identification, and correlation analysis between usage patterns and urban activity rhythms. These insights are valuable for understanding how shared bicycles integrate with broader urban transportation systems and daily activity patterns.

Demand forecasting applications. Although the dataset covers only a limited period, it provides a useful empirical basis for testing and demonstrating demand prediction approaches, such as the BPNN model. These results illustrate the methodological feasibility of short-term demand forecasting and can inform future applications when combined with extended datasets.

The high temporal resolution and spatial granularity support development of sophisticated forecasting approaches that can predict demand at fine geographical and temporal scales, enabling real-time operational optimization and dynamic resource allocation strategies.

Service area optimization. The regional clustering methodology proposed in this work provides a methodological basis for defining potential operational zones and assessing service balance across the city. While not directly implementable at a large operational scale due to the dataset's temporal limitation, it demonstrates a framework that can support service area optimization when longer-term data become available.

The clustering results provide practical guidance for fleet management decisions, maintenance scheduling, and infrastructure investment planning, supporting evidence-based operational optimization strategies.

Integration with other transportation data. This dataset can be effectively integrated with complementary transportation datasets including public transit usage data, taxi service records, and pedestrian traffic counts to enable comprehensive analysis of multimodal transportation patterns. Such integrated analysis supports improved understanding of transportation system interactions and can inform comprehensive urban mobility planning approaches.

The precise temporal and spatial resolution of this dataset makes it particularly suitable for integration with other high-resolution urban data sources, enabling sophisticated analysis of transportation mode interactions and transfer behaviors.

Data availability

The complete dataset is publicly available in the Zenodo repository at <https://doi.org/10.5281/zenodo.17415891>. The repository contains all data files in CSV format with UTF-8 encoding, including order data (220,675 trip records), trajectory data (2,849,243 GPS tracking points), electronic fence data (14,071 parking zone records), weather data, and preprocessing algorithms. Comprehensive documentation including README file, data dictionary, and technical specifications are provided to facilitate data reuse.

Code availability

The code used in the dataset processing sessions and model building is provided at <https://doi.org/10.5281/zenodo.17415891>.

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Author contributions

Jianrong Cai, Qiang Wen, and Tao Chen contributed to the conceptualization and design of the study. Lian Xie and Jie Yu performed the data collection and analysis. Yang Liu contributed to the methodology and interpretation of results. All authors were involved in writing and reviewing the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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